



Technology Stack

- **Mobile App:** React Native (iOS/Android), Reanimated 3, MapLibre GL radar
- **Backend / API:** Python 3.11 ([FastAPI](#) Async), WebSockets ([WSS](#)), REST
- **AI / ML & Graph DL:** PyTorch Geometric ([ST-GAT](#) Corridor Attention), ONNX Runtime
- **Physics Kinematics:** Newton-Davis tractive engine, grade, air drag & RK4 ODE solver
- **Data Streaming:** Apache Kafka (100k+ event/s KRaft), Redis 7.2 ([Pub/Sub](#) cache)
- **Spatial & Time-Series:** PostgreSQL 16 + [PostGIS](#) (1D rail chainage LRS), TimescaleDB
- **Railway Ingestion:** BEL RTIS (Navic 30s GPS), CRIS FOIS freight, COA logs, e-Caution TSR
- **Security & Hosting:** OAuth 2.0 / JWT, AES-256 GCM, Docker, Kubernetes on RailCloud / NIC

100% ZERO-HARDWARE Taps statutory Indian Railways GPS & CRIS feeds. Zero engine retrofits.

Powered By

Python

PyTorch

React Native

Kafka

Redis

PostgreSQL

PostGIS

FastAPI

Docker

Kubernetes

ISRO NavIC

CRIS / IR

TECHNICAL APPROACH

End-to-End User & System Flow

Traveler & Controller Flow

1. User Opens GATI-SETU App
Traveler on train/station opens clean "Where Is My Train" style app

MOBILE / WEB

2. Search Train # or GPS Geofence
Enter #12302 (Rajdhani) or GPS auto-detects nearest railway corridor

AUTO-DETECT

3. 30s BEL RTIS Navic Satellite Ingress
Ingests live locomotive MSS transponder GPS + CRIS COA block state

ZERO-HARDWARE

GPS Signal Lost in Tunnel/Gorge?
Line-of-Sight Satellite Handshake

YES: Newton Dead-Reckoning
RK4 solver computes displacement

NO: PostGIS 1D Rail-Snap
Snaps GPS to track chainage KM

5. Dynamic Kinematics + PyG ST-GAT Core
WAP-7 tractive curve physics fused with spatio-temporal graph attention

DUAL AI ENGINE

Schedule Delay > 5 Mins or Preceding Conflict?
Section Capacity & Train Precedence Evaluation

Nominal: 4-Aspect Green Wave
130 km/h corridor lock; on time

Conflict: Rule 401 Siding
Loops freight; Express recovers 22m

7. Calibrated ETA [P10-P90] & Coach Guide
Live arrival window, platform #4*, coach formation & causal reason badge

LIVE DISPLAY

8. 1-Tap Cab & Metro Sync • Journey Done
Pre-timed Ola/Uber dispatch to platform touchdown; zero wait, zero panic

SEAMLESS

Core Architecture & Live Working Prototype

Tested Live • Zero Engine Hardware Required

Statutory Ground Feeds (100% Zero-Hardware)

INGESTION LAYER

Taps existing **BEL RTIS Navic** 30s satellite GPS transponders on locos + **CRIS COA** interlocking block occupancy and **e-Caution TSR** temporary speed restrictions.

PostGIS 1D LRS & Newton-Davis Kinematics

PHYSICS ENGINE

Snaps raw 2D WGS84 GPS drift to 1D track chainage KM. Runge-Kutta 4th-order ODE solver models tractive effort curves (WAP-7 6350HP), brake deceleration & 1:120 ruling gradients.

$$a(t) = [F_{trac}(v) - R_{davis}(v) - F_{grade}] / M_{effective}$$

PyG Spatio-Temporal Graph Attention (ST-GAT)

AI CORE

Models the entire railway corridor as a dynamic spatio-temporal graph. Multi-head cross-train attention captures downstream ripple delays across interconnected stations up to 150 km ahead.

$$\sigma_{ij} = \text{Softmax}(\text{LeakyReLU}(a^T [Wh_i || Wh_j]))$$

Inductive Conformal Uncertainty Calibration

CERTIFIED [P10-P90]

Replaces unreliable single-point arrival estimates with a certified 90% non-parametric confidence window (**23:48 - 23:46**), resilient to fog, crossings & signal halts.

Multi-Stakeholder Operational Synchronization

DISPATCH ENGINE

Rule 401 Siding Engine automatically loops low-priority freight to sidings (recovering ~22m delay for express trains) • Syncs with **Station Master CIDS** platform LED and audio displays.

Continuous Closed-Loop Operational Feedback: BEL RTIS updates satellite coordinates every 30s • Auto-recalibrates ST-GAT corridor graph & Conformal quantiles • 100% Zero-Hardware

Statutory IR Compliant



Working Mobile Passenger Prototype